Novia One.

FMP · MODULE 4 · WORKBOOK

PRACTICAL EVIDENCE · EQUITY SECURITIES

Workbook, answered.

Model answers to all **10 questions (150 marks)**, matched to the lecturer's marking style: bullet points with examples for list questions, prose for conceptual ones. Answers verified against the lecturer's memo are tagged MEMO.

Note: study/model answers to learn from. The workbook is assessed as **your own work** (pass mark **65%**) — reword in your own voice for submission.

MARK ALLOCATION AT A GLANCE

QUESTION	TOPIC	MARKS
Q1	Equity fundamentals & capital structure	15
Q2	Equity income, share types, listed/unlisted, DRs	15
Q3	Determinants of optimal capital structure	15
Q4	Debt versus equity	10
Q5	Primary & secondary markets, IPOs, listing	30
Q6	Listing reasons, methods, rights issue, risks	15
Q7	Gordon Growth model — shortcomings	5
Q8	Market capitalisation (FSR vs Nedbank)	10
Q9	Valuation techniques by sector	10
Q10	Dual-class shareholding case study	25
Total		**150**

ANSWER STYLE List questions use a lead-in + bold-label bullets (with examples); conceptual and discursive questions are answered in prose, often with a "However..." second paragraph. **MEMO** = matched to the lecturer's actual answer.

QUESTION 1 · EQUITY FUNDAMENTALS & CAPITAL STRUCTURE [15] MEMO

1.1 Main advantages of raising equity capital for a company? (3)

The main advantages of raising equity capital include no additional financial burden from interest payments, no obligation to repay the capital, no mandatory fixed payments (dividends are discretionary), and no maturity date — all of which give the company valuable operational and financial flexibility. In addition, bringing in large or institutional investors can provide business expertise, resources and valuable contacts, while a stronger equity base lowers gearing and supports future borrowing.

1.2 Factors that influence the optimal capital structure? (3)

The optimal capital structure of a company is influenced by factors such as the corporate lifecycle, the tax-deductibility of interest, business risk, bankruptcy (financial-distress) risk, agency costs, and lender risk appetite. Together these factors determine the ideal balance of debt and equity — the mix that minimises the company's cost of capital and maximises its market value.

1.3 Purpose of equity in the capital markets? (3)

The purpose of equity in the capital markets includes raising capital for businesses through share offerings, providing a marketplace for buying and selling shares, protecting investors against fraud, ensuring regulatory compliance, providing real-time price information, maintaining an orderly market, and offering investors a share in company profits.

1.4 Role of equity in a company's capital structure? (3)

Equity in a company's capital structure represents ownership in the business through common and preferred stock, together with retained earnings. It offers advantages such as no obligation for repayment or interest payments and greater operational flexibility, but it can also lead to a loss of control and to profit-sharing with new shareholders through dilution.

1.5 How does the Debt-to-Equity (D/E) ratio help assess financial risk? (3)

The Debt-to-Equity (D/E) ratio measures the financial risk of a company by comparing its total debt to the value of its shareholders' equity. A higher D/E ratio indicates greater reliance on borrowed funds and therefore greater financial risk — more fixed obligations and higher exposure to default in a downturn — whereas a lower ratio indicates a more conservatively financed, lower-risk company. For example, a D/E ratio of 2.5 means the company has R2.50 of debt for every R1.00 of equity.

QUESTION 2 · EQUITY INCOME, SHARE TYPES, LISTED/UNLISTED, DRS [15]

2.1 Two primary forms of income from investing in equities? MEMO (3)

The two primary forms of income generated from investing in equities are:

- **Capital growth:** the increase in the value of shares when their market price rises — for example, if a share bought for R100 rises to R110, the investor gains 10% in capital growth.
- **Dividends:** payments made by a company to its shareholders out of its profits, typically paid on a regular basis, offering a steady income stream.

2.2 How does investing in preference shares differ from ordinary shares? MEMO (3)

Investing in preference shares differs from investing in ordinary shares in several ways:

- Preference shareholders receive their dividends **before** ordinary shareholders and typically have a **fixed** dividend rate.
- Ordinary shareholders have **voting rights** and participate in company governance, whereas preference shareholders usually do **not** have voting rights unless specified.
- In the event of **liquidation**, preference shareholders have **priority** over ordinary shareholders in receiving their share of the company's assets.

2.3 Significance of voting rights for ordinary shareholders? MEMO (3)

Voting rights for shareholders in ordinary shares are significant because they allow investors to participate in key decisions within the company. This includes voting on issues such as the election of directors, mergers, and the appointment of auditors. Shareholders can exercise their voting rights at the annual general meeting, either in person or via proxy. These voting rights provide shareholders with influence over the company's management and strategic direction.

2.4 Main differences between listed and unlisted equity securities? (3)

Listed and unlisted equity securities differ in several ways:

- **Market & trading:** listed shares trade on a licensed exchange (a public market) with continuous trading, while unlisted shares trade over-the-counter through dealers or market makers (a private market) with irregular trading.
- **Regulation & transparency:** listed shares are regulated under the Financial Markets Act with transparent pricing; unlisted shares have little regulation and opaque pricing.
- **Liquidity & risk:** listed shares are liquid and lower-risk, whereas unlisted shares are illiquid, carry wide bid-ask spreads and the potential for price manipulation, making them riskier.

2.5 Role & characteristics of a depository receipt (DR)? MEMO (3)

A depository receipt (DR) is a security that allows investors to hold shares of foreign companies without directly trading in foreign markets. It represents an economic interest in a foreign company and is traded on a local exchange. DRs offer convenience and cost-efficiency compared to direct foreign stock purchases, while allowing investors to diversify their portfolios. They provide the same economic, corporate and voting rights as owning the foreign shares directly. However, DRs come with risks such as lower liquidity and exposure to currency fluctuations.

QUESTION 3 · DETERMINANTS OF OPTIMAL CAPITAL STRUCTURE [15]

3.1 **How does a company's lifecycle stage influence its optimal capital structure?** (3)

A company's lifecycle stage strongly influences its optimal capital structure. Start-up and growth companies have uncertain cash flows and few tangible assets, so they rely mainly on equity such as venture capital and carry little debt. Mature companies, with stable cash flows and a solid asset base, can support more debt and benefit from the interest tax shield. Companies in decline tend to reduce debt or return capital, since over-gearing at this stage sharply raises the risk of financial distress.

3.2 **Why is business risk a determinant of optimal capital structure?** (3)

Business risk — the variability of a company's operating earnings — is a key determinant because taking on debt layers financial risk on top of it. A firm with high business risk that also carries high debt compounds its total risk and raises the probability of financial distress. Such firms should therefore use less debt to keep total risk manageable, whereas firms with stable, predictable earnings can safely carry more.

3.3 **Why does increasing leverage initially raise firm value, and what risk offsets this?** (3)

Increasing leverage initially raises firm value because debt is cheaper than equity and interest is tax-deductible; this tax shield lowers the weighted average cost of capital and lifts the firm's value. The offsetting risk is financial-distress or bankruptcy risk: as debt rises, so does the probability and cost of distress. Beyond the optimal point these distress costs outweigh the tax-shield benefit and firm value begins to fall — the essence of trade-off theory.

3.4 **What role does lender risk-appetite play for early-stage companies?** (3)

Lender risk appetite is critical for early-stage companies because they are inherently risky, with no track record, few assets and uncertain cash flows. Lenders are therefore reluctant to advance funds and demand high interest rates or collateral that the company cannot provide. This limited willingness to lend effectively caps the amount of debt available, forcing early-stage firms to rely on equity such as angel or venture-capital funding.

3.5 **How do agency costs affect capital-structure decisions?** (3)

Agency costs affect capital-structure decisions because of conflicts between the parties involved. The agency cost of equity arises when managers misuse free cash flow, for example through empire-building; taking on debt imposes discipline by committing that cash to fixed repayments. The agency cost of debt arises when shareholders take excessive risk or underinvest at the lenders' expense, which prompts lenders to impose covenants and charge higher rates. The capital structure is set to balance these opposing costs.

QUESTION 4 · DEBT VERSUS EQUITY [10] MEMO

4.1 Primary difference between debt and equity (obligations & ownership)? (2)

The primary difference is that debt securities represent a liability for the issuing company, which must repay the principal and pay interest at specified intervals. In contrast, equity securities do not impose a repayment obligation on the company; instead, equity shareholders are owners with a residual claim on the company's assets after all liabilities have been paid. Equity investors are not guaranteed periodic payments, but they seek a total return from capital appreciation and dividends.

4.2 How do equity investors' objectives differ from debt investors'? (2)

Equity investors are seeking a total return, which includes both capital appreciation (the increase in the market price of the shares) and dividend income, and they invest in the expectation that the company's management will make decisions that maximise shareholder wealth. Debt investors, on the other hand, are primarily interested in receiving interest income from the debt securities they purchase, and they typically expect to hold those securities until maturity, when the principal is repaid.

4.3 Characteristics of preference shares vs ordinary shares? (2)

Preference shares (preferred stock) have characteristics that differ from ordinary shares. Preference shareholders have a fixed dividend entitlement that takes priority over dividends to ordinary shareholders, and in the event of liquidation they are paid before ordinary shareholders but after debt holders. Unlike ordinary shareholders, preference shareholders typically do not have voting rights; however, they receive a more stable income stream than ordinary shareholders, who may receive variable dividends depending on the company's profitability.

4.4 Why are equity investors owners, and how does this affect their claims? (2)

Equity investors are considered owners because they hold shares in the company, which gives them ownership rights. This ownership provides them with a claim on the company's assets after all liabilities have been settled, meaning they have a residual claim — once the company's debts are paid off, equity shareholders are entitled to any remaining assets or profits. However, unlike debt investors, equity investors do not have guaranteed returns and are exposed to more risk, since their returns depend on the company's performance and market conditions.

4.5 Main advantage for a company issuing equity instead of debt? (2)

The main advantage for a company in issuing equity securities is that it does not incur any obligation to repay the capital raised or to make periodic interest payments, unlike debt securities. This provides the company with more financial flexibility, as there are no fixed repayment schedules, and issuing equity can improve the company's debt-to-equity ratio, making it potentially easier to secure future financing. The trade-off, however, is that issuing equity dilutes existing shareholders' ownership and control.

QUESTION 5 · PRIMARY & SECONDARY MARKETS, IPOS, LISTING [30]

5.1 Main difference between primary and secondary equity markets, and their functions? (5)

The main difference is that the primary market is where a company issues new shares and raises capital, for example through an IPO or a seasoned offering, serving the function of capital formation, while the secondary market is where investors trade existing shares among themselves, with no new capital flowing to the company. The secondary market performs the functions of providing liquidity, price discovery and continuous, fair valuation. The two are closely linked, because investors are willing to buy in the primary market only because the secondary market lets them sell later.

5.2 Explain the IPO process and how it helps a company raise capital. MEMO (5)

An Initial Public Offering (IPO) is the process by which a company offers its shares to the public for the first time, effectively "going public." The company works with underwriters, typically investment banks, who assist in pricing the shares, marketing the offering and facilitating the sale. Through the IPO process the company sells newly issued shares to the public, raising capital that can be used for purposes such as expansion, reducing debt or improving operations. The IPO is a significant way for a company to access large amounts of capital from public investors and to grow and gain visibility in the market, and by listing it also gains credibility that can help attract further investment and partnerships. However, the IPO process is complex and costly, involving significant regulatory requirements, underwriting fees and the responsibility of meeting ongoing public disclosure standards.

5.3 Advantages and disadvantages of listing on the JSE? MEMO (5)

Listing on the Johannesburg Stock Exchange (JSE) offers several advantages for a company. The primary benefit is access to a broader pool of capital, as the JSE attracts both local and international investors. By listing, a company can raise significant funds for growth, reduce its debt burden and enhance its reputation. The JSE also provides liquidity for shareholders, allowing them to buy and sell shares with ease, and it offers transparency and regulatory oversight, which can increase investor confidence.

However, there are also disadvantages. The process can be expensive due to underwriting costs, listing fees and other administrative expenses. Public companies are subject to greater scrutiny, as they must comply with rigorous financial reporting requirements and regulatory obligations, and this increased accountability can be challenging, especially for smaller companies or those without a strong track record. Moreover, listing on the JSE may dilute ownership for existing shareholders, as new shares are issued to the public.

5.4 Difference between an IPO and a seasoned equity offering (SEO), and the effect of each? MEMO (5)

An Initial Public Offering (IPO) occurs when a company offers its shares to the public for the first time, transitioning from private to public ownership; this allows it to raise capital by selling new shares, typically used for business expansion, debt reduction or other corporate purposes, and it also gives the company greater visibility, credibility and access to future capital. In contrast, a seasoned equity offering (SEO) involves an already-public company offering additional shares, either through a cash offer (selling new shares to the public) or a rights offer (offering shares to existing shareholders), to raise further capital after going public — often to fund acquisitions or pay down debt. While an IPO brings new capital into the company and raises its public profile, it can result in ownership dilution for existing shareholders; an SEO can also dilute the value of existing shares, but it typically faces less risk since the company is already public and the market price is well-established.

5.5 Role of underwriters in listing equity, and the risks they face? MEMO (5)

Underwriters are financial institutions, typically investment banks, that assist companies in the process of issuing new securities, whether in an IPO or a seasoned equity offering. They play a crucial role in determining the price of the securities, managing the sale and ensuring it is successfully marketed to investors. Underwriters often purchase the securities from the issuing company and then sell them to the public or institutional investors; by doing so they assume the risk of any unsold shares, guaranteeing that the company receives the agreed amount of capital from the offering. The risks for underwriters are significant: if the securities are priced incorrectly or market conditions are unfavourable, they may not be able to sell all the securities at the expected price, resulting in a financial loss, and they must also comply with all regulatory requirements, which can involve substantial legal and administrative costs. However, underwriters are compensated for their services with fees, usually a percentage of the funds raised.

5.6 Advantages of secondary markets for investors and companies, and their key functions? MEMO (5)

Secondary markets, such as stock exchanges, provide several advantages for both investors and companies. For investors, the primary advantage is liquidity: they can buy and sell securities quickly and at transparent prices, making it easier to convert investments into cash and allowing easier entry and exit, which makes investing in equities more attractive. For companies, secondary markets provide valuable price discovery, reflecting the market value of their securities based on investor demand and sentiment; although the company does not benefit directly from secondary-market transactions, a liquid and active market enhances the attractiveness of its shares and can boost its market value, while increasing visibility and improving its reputation and market standing. The key functions of secondary markets include providing liquidity, enabling price discovery, offering transparency and supporting primary-market transactions by making securities more attractive to investors — helping the equity securities market operate efficiently and effectively.

QUESTION 6 · LISTING REASONS, METHODS, RIGHTS ISSUE, RISKS [15] MEMO

GIVEN The JSE offers secure primary & secondary markets; **Company ABC** plans to list on the JSE.

6.1 Reasons for seeking a stock-market listing? (6)

There are several reasons why companies seek a stock-market listing:

- It gives them **access to a wider pool of finance**, making it easier to obtain funds.
- Shares become **more marketable**, which in turn increases the share price.
- It brings an **enhanced public image** and more exposure in the media.
- A listing can be used by the **original owners as an exit strategy** or to realise some of their shareholding, and is sometimes seen as a step to the next level where a stronger management team can be recruited.
- Original owners may wish to **sell their shareholding to obtain funds** for other projects.
- Listed companies find it **easier to grow by acquisition** than unlisted companies.

6.2 Describe two methods of obtaining a listing. (2)

Two methods that can be used to obtain a listing are:

- **An offer for sale:** investors tender for shares, which are then allocated to them.
- **A placing:** a number of specific investors, such as pension funds or business associates, are invited to take up the shares.

6.3 A listed company raises R2bn in equity (rights issue). If you do NOT follow it, what happens to your holding? MEMO **(3)**

If you do not follow the rights issue:

- It **dilutes your percentage of the company owned.**
- This is a **bad option**, because the shareholder experiences a dilution in share value without the compensation they could have received by selling the rights.

6.4 Risks of trading in the equity market? MEMO **(4)**

The risks of trading in the equity market include:

- **Interest rate risk** — changing rates affect share prices and the cost of capital.
- **Currency risk** — exchange-rate movements affect returns, especially on foreign exposure.
- **Commodity risk** — commodity-price moves affect resource-linked shares.
- **Bankruptcy risk** — the company may fail, and equity holders rank last.

QUESTION 7 · GORDON GROWTH MODEL [5]

GIVEN Blue-chip telecoms: dividend = **820c**, growth **g = 1%**, required return **r = 8%**.

7.1 Shortcomings of the Gordon Growth model to keep in mind. **(5)**

Context (the model's output): Value = $D_0(1+g)/(r-g)$ = 820 × 1.01 ÷ (0.08 − 0.01) = 828.2 ÷ 0.07 ≈ **11 831 cents (about R118.31)**.

The investor should, however, keep several shortcomings in mind:

- It assumes a single, **constant growth rate in perpetuity**, which is unrealistic for real companies.
- It is **extremely sensitive to its inputs** — a small change in the required return or the growth rate produces a large change in the calculated value.
- It **breaks down when g ≥ r**, giving a negative or meaningless result.
- It applies only to **mature, stable, dividend-paying companies** — not to high-growth or non-dividend payers.
- It relies heavily on **estimated inputs** and ignores other value drivers and qualitative factors.

QUESTION 8 · MARKET CAPITALISATION — FSR VS NEDBANK [10] MEMO

GIVEN FirstRand (FSR) share price **R47.33**; Nedbank (NED) share price **R206.50**. Does NED's higher price mean it is more valuable?

8.1 Describe market capitalisation using these two companies. (4)

Market capitalisation is the number of shares in issue multiplied by the share price. The share price on its own gives no indication of how much equity a company has, so the R47.33 and R206.50 quoted for FirstRand and Nedbank cannot be compared directly — you must multiply each company's share price by its number of shares in issue. A company with a lower share price but far more shares in issue can therefore have a much larger market capitalisation.

8.2 Which of the two has the greater equity value? (2)

FirstRand (FSR) has the greater equity value. Although Nedbank trades at a higher share price, FirstRand has far more shares in issue, which gives it a larger market capitalisation and therefore greater equity — showing that a higher share price does not mean a more valuable company.

8.3 Describe four benefits of being a listed company. (4)

Four benefits of being a listed company are:

- **Liquidity:** shares can be readily bought and sold on the exchange.
- **Regulatory / audit controls:** the company is subject to exchange regulation and audit requirements, which improves its credibility.
- **Easily obtainable valuation:** the market provides a transparent, real-time valuation of the company.
- **Guaranteed settlement:** trades are cleared and settled through the exchange's clearinghouse.

QUESTION 9 · VALUATION TECHNIQUES BY SECTOR [10]

9.1 Give an example of a listed company in that (telecommunications) sector. **(1)**

An example of a listed company in the telecommunications sector is MTN Group; other examples include Vodacom Group and Telkom, all listed on the JSE.

9.2 List three valuation techniques, the companies/sectors each suits, and why. **(9)**

TECHNIQUE	SUITED TO	REASON
Dividend Discount Model (DDM / Gordon Growth)	Mature, stable dividend payers — telecoms, banks, utilities	Regular, predictable dividends make the present value of dividends meaningful.
Discounted Cash Flow (FCFF / FCFE)	Firms with no or irregular dividends but predictable cash flows — industrials, growth firms	Values the intrinsic cash-generating ability directly, even without dividends.
Relative valuation (P/E, EV/EBITDA)	Firms with comparable listed peers and positive earnings — retail, consumer	Gives a quick, market-based benchmark against similar companies.

An asset-based or NAV model also suits asset-heavy firms such as property/REITs and investment holding companies, where value equals total assets minus total liabilities.

QUESTION 10 · DUAL-CLASS SHAREHOLDING CASE STUDY [25]

GIVEN **Naspers** (N shares = 1 vote; unlisted A shares = 1 000 votes, held by management) and **Alphabet/Meta** (multi-class: high-vote insider shares + low/no-vote public shares) — dual-class structures that concentrate control with founders/insiders.

10.1 How do these structures impact corporate governance? (3)

These dual-class structures concentrate control in the hands of insiders and breach the principle of one share, one vote. They weaken the board's accountability to ordinary shareholders and can entrench management. The trade-off is that, while they provide stability and a long-term vision, they also reduce the checks and balances that normally protect minority shareholders.

10.2 Could concentrated insider voting compromise accountability to shareholders? (3)

Yes. Because high-voting shareholders can override ordinary shareholders, they are able to resist shareholder activism and act with fewer checks. Ordinary N or Class A shareholders can never outvote the insiders, no matter how many shares they hold, which materially reduces management's accountability to the broader shareholder base.

10.3 How might these structures affect an investor's decision to buy? (3)

These structures can deter some investors, who see the limited say and the governance risk as reasons to avoid the shares, while others are willing to accept them where management has a strong track record, as with Naspers's transformational investment in Tencent. Ultimately the decision depends on the investor's trust in management and on the price or valuation on offer.

10.4 Do investors perceive higher risk from diminished voting power, and how is it reflected in price/valuation? (3)

Yes. Diminished voting power is viewed as a governance risk, and this is often reflected in a governance discount, where the low- or no-vote shares — such as Naspers's N shares or Alphabet's Class C shares — tend to trade at a discount to the high-vote shares. Investors demand a higher required return to compensate for the weaker rights, and this translates into a lower valuation.

10.5 Would the companies make different strategic decisions if voting power were more equal? (2)

Quite possibly. With votes distributed more equally, management would face greater short-term shareholder pressure and might make less bold, less long-term decisions. Transformational bets such as Naspers's early investment in Tencent might not have been made under that kind of pressure.

10.6 How do these structures serve as a defence against hostile takeovers? (2)

Because the insiders' concentrated votes mean that an acquirer cannot gain control simply by buying up the publicly traded shares, the structure acts as an effective shield against hostile takeovers.

10.7 Other mechanisms to maintain control without multi-class shares? (2)

Yes — companies can use mechanisms such as poison pills, staggered or classified boards and supermajority voting provisions, as well as golden shares, shareholder agreements or a pyramid/holding-company structure, to maintain control without resorting to multi-class shares.

10.8 How do these structures affect market perception of transparency & governance? (2)

They can signal weaker governance and transparency, which is a reputational concern and has led some stock-market indices to exclude dual-class companies. However, consistently strong performance can offset this concern and keep investors comfortable with the structure.

10.9 How sustainable are these structures in the long run? (3)

Their sustainability is mixed. They can persist for as long as the founders continue to deliver strong performance and retain their high-vote shares, but pressure builds over time through shareholder activism, index exclusion and succession when founders eventually exit. Sunset clauses that automatically convert the shares to one-share-one-vote are increasingly common, and the structures may prove unsustainable if performance falters.

10.10 Could these companies alter their structures in future? (2)

Yes — a company might alter its structure if a founder exits, dies or dilutes below a key threshold, if investor, regulatory or index pressure becomes strong enough, if a sunset clause triggers a conversion to one-share-one-vote, or if a large capital requirement forces it to issue broader equity.

SUBMISSION REMINDER Write your final answers in your **own words**, number them to match the workbook, and put your name, surname & ID on every page. Pass mark = **65%**.